

Seminar Arbeit

From the Gai Kapadia model to it's empirical verification

—

Are more connections increasing the stability of the system ? From the Gaussian point of view to the Power Law

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1 Introduction

The modern financial landscape is defined by an intricate web of highly interconnected institutions. While this density of connections is theoretically designed to bolster stability through risk-sharing, it also creates dangerous pathways for distress to spread across the global economy. History has repeatedly demonstrated the devastating ripple effects that a single localized shock can trigger, making the study of systemic risk a top priority for researchers and regulators alike.

In this paper, I will build upon the foundational work presented by Clément Landry regarding financial networks. While Clément focused on the structural representation of these systems, my objective is to dive deeper into the **Gai & Kapadia model** [5] to evaluate its mechanics and, more importantly, its limitations.

The fundamental question we aim to answer is: does increased connectivity truly enhance the stability of the system? We will demonstrate that the original Gai-Kapadia framework is largely built on the assumption of a **Gaussian distribution** (Erdős-Rényi type networks). However, empirical evidence from real-world interbank markets suggests that these networks actually follow a **Power Law** (Scale-free networks).

Throughout this report, we will explore how this shift from a Gaussian perspective to a Power Law reality radically changes the system's resilience. After a brief overview of the necessary graph theory tools, we will analyze the direct contagion model and use numerical simulations to test its limits. Finally, we will introduce illiquidity effects and discuss how financial policies, such as Capital Buffers, must account for these non-linear "robust-yet-fragile" transitions to prevent global collapse.

2 Network and Graph Theory

2.1 The “Why”: From Isolation to Interdependence

In traditional financial models prior to 2008, banks were often treated as isolated agents. If one institution failed, it was viewed as an independent event. However, the collapse of Lehman Brothers demonstrated that **systemic risk is a collective phenomenon**.

Banks are not isolated islands; they borrow, lend, and share risks through complex webs of contracts. A network is the natural language to represent these interdependencies. Because risk propagates through these connections rather than in isolation, a small shock can cascade into a global crisis.

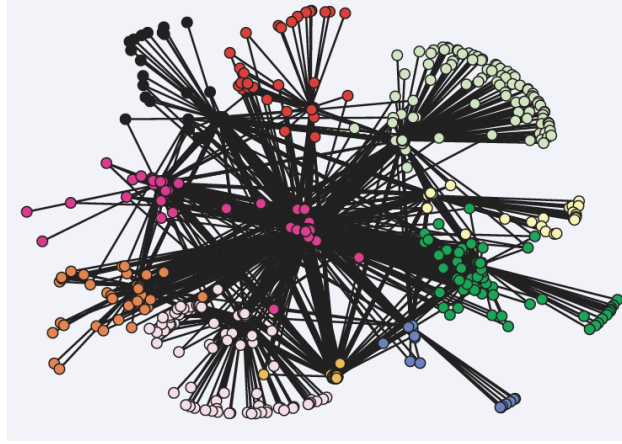


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2.2 The “What” and “How”: Graph Theory Basics

We describe a financial system using a **graph** $G(V, E)$, where:

- **Vertices (Nodes) V :** Represent economic agents (banks, insurance companies, or households). $k_i^{\text{in}} = \sum_j A_{ji}$
- **Edges (Links) E :** Represent the relations between them (loans, ownership, or common investments). $k_i^{\text{out}} = \sum_j A_{ij}$

A fundamental tool to describe these graphs is the **adjacency matrix** A :

$$A_{ij} = \begin{cases} 1 & \text{if there is a link from node } i \text{ to node } j \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

$$V = \{1, 2, 3, 4, 5\}$$

$$E = \{1 \rightarrow 2, 3 \rightarrow 4, 5 \rightarrow 2, 4 \rightarrow 5, 3 \rightarrow 5\}$$

2.3 Types of Financial Relations

Financial links vary in nature and complexity. While we often focus on single-layer graphs for mathematical simplicity, it is important to recognize the different forms these relations take:

- **Directed vs. Undirected:** In a **directed** graph (e.g., a loan), the link from i to j is not the same as j to i . In an **undirected** graph (e.g., a partnership), the relation is mutual, making the matrix A symmetric ($A_{ij} = A_{ji}$).
- **Weighted Graphs:** If the intensity of the connection matters, we assign a weight ω_{ij} to each edge. In finance, this weight typically represents the monetary value of a debt or the size of an exposure.
- **Bipartite and Co-occurrence Networks:** A **bipartite network** involves two distinct sets of nodes, such as Banks and Assets. Two banks may have no direct contract but are linked because they hold the same asset. We can project this into a single-layer **co-occurrence network** to uncover hidden dependencies between portfolios.
- **Multiplex Networks:** Real-world agents are connected by multiple types of links simultaneously (e.g., debt, ownership, and derivatives). These are modeled as **multi-layer graphs**, where each layer represents a different type of relationship.

2.4 Quantitative Tools for Contagion

To predict how a shock will spread, we characterize the network's structure using several key metrics:

Degree Distribution (P_k): The probability that a node has k connections. A distribution concentrated on a few high-degree “hubs” indicates a system where the failure of a single major bank can immediately affect the entire network.

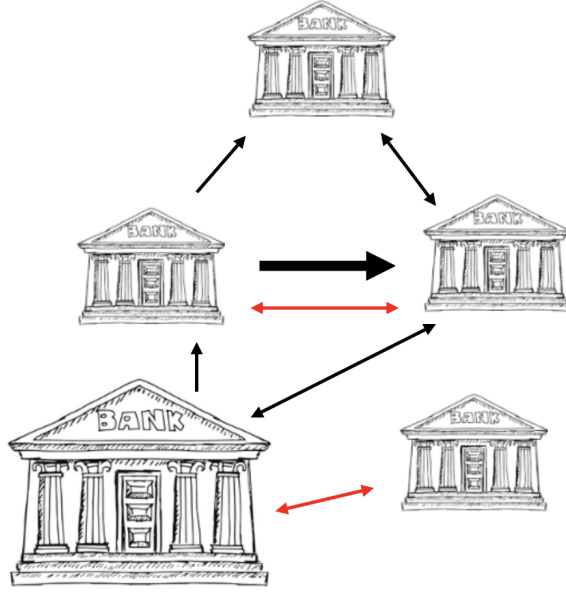


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Average Degree (z): In a directed graph, the average in-degree equals the average out-degree:

$$z = \sum_{j,k} j \cdot P_{jk} = \sum_{j,k} k \cdot P_{jk} \quad (2)$$

Clustering (C): This measures the probability that the neighbors of a node are also neighbors of each other. High clustering can amplify contagion by trapping and intensifying shocks within a tightly-knit group of institutions.

Path Length and Small-World Property: The **average path length** $\langle \ell \rangle$ is the average number of steps to go from one node to another. In “small-world” networks, $\langle \ell \rangle$ is very low, meaning a local shock can become global almost instantaneously.

2.5 A Physics Perspective

As physics students, we find this framework particularly enriching because it allows for direct parallels with statistical mechanics. Financial contagion can be viewed as a **phase transition** or a **percolation problem**.

Just as a single particle’s state in a lattice can trigger a change in the entire system’s equilibrium, a single bank’s distress can shift the state of the financial ecosystem. Our goal is to move from monitoring individual institutions to understanding the collective dynamics that ensure global financial stability.

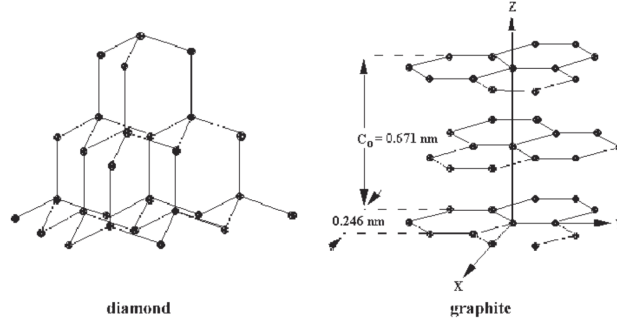


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3 Gaussian vs. Power Law: The Nature of Shocks

Before diving into networks and contagion, we need to understand the nature of the shocks to which the system is exposed. The architecture of a network matters, but so does the size of the "spark" that starts the fire.

The probability distribution of shock sizes determines whether extreme events are treated as impossible "Black Swans" or as structural certainties.

3.1 Two Worlds: Mediocristan vs. Extremistan

In the study of financial stability, we generally distinguish between two regimes:

1. **The Gaussian World (Thin Tails):** Most observations cluster around the mean. Large shocks are exponentially suppressed. In this world, a 10σ event (a "10-standard deviation" shock) would not happen even once in the history of the universe.
2. **The Power-Law World (Heavy Tails):** The distribution decays slowly. Large shocks are rare but **expected**. Here, the variance can be infinite, meaning the system has no "typical" scale for disasters. Extreme events are not negligible—they are structurally present.

3.2 Mathematical Comparison

To compare these two regimes, we look at their Probability Density Functions (PDFs):

The form of the distribution is not a statistical detail; it is the primary factor determining the stability of the system. If we assume a Gaussian world when the reality is Power-Law, we will consistently underestimate systemic risk.

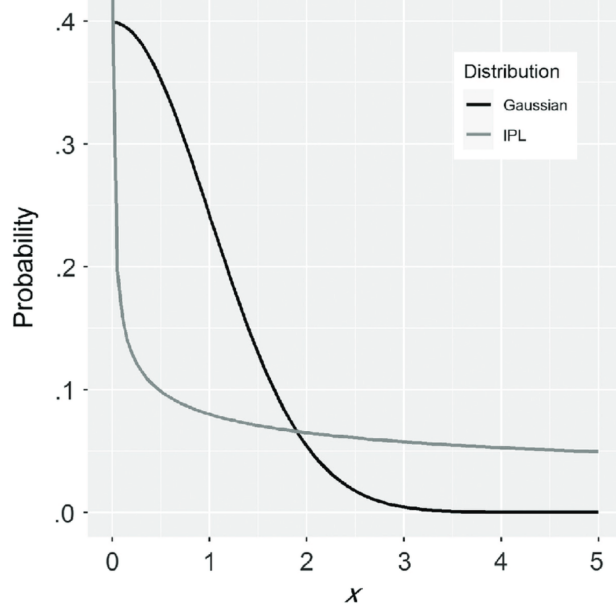


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Feature	Gaussian (Normal)	Power Law (Pareto)
Equation	$f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$f(x) = Cx^{-\alpha}$
Mean ($E[X]$)	μ	$\frac{\alpha x_{min}}{\alpha-1}$ (for $\alpha > 1$)
Tail Decay	Exponential (Rapid)	Polynomial (Slow/Heavy)
Stability	High (Predictable)	Low (Prone to "Black Swans")

Table 1: Comparison of Gaussian and Power Law distributions.source veritasium video

3.3 Implicit Gaussianity in the Gai-Kapadia Model

In the seminal Gai-Kapadia (GK) model, the network is often modeled as an **Erdős-Rényi graph**. In this framework, each pair of nodes is connected with a probability p . This results in a Poisson degree distribution which, for a large number of nodes, behaves like a Gaussian distribution.

In such a system:

- No bank has an "absurd" or disproportionate number of connections (no massive hubs).
- Interbank exposures are normalized and i.i.d. (independent and identically distributed).
- Under the **Central Limit Theorem (CLT)**, the total loss becomes highly predictable.

As we will prove in the next section, the GK model operates in a regime where losses are "self-averaging." While this makes the mathematics elegant, it may fail to capture the "fat-tailed" reality of modern financial markets where a single node's failure can outweigh the combined stability of the rest of the network.

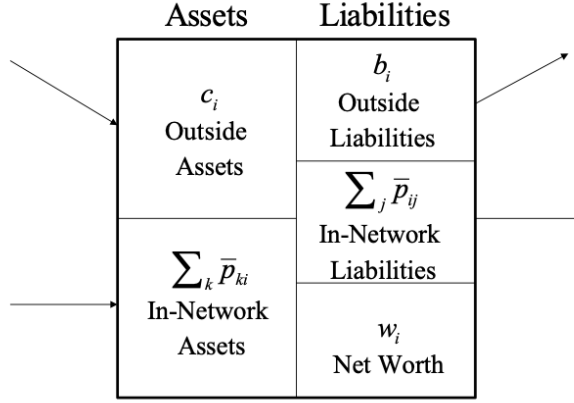


Figure 5: Enter Caption

4 The Gai-Kapadia Model: Robust yet Fragile

The seminal work by Gai and Kapadia (2010) provides a cornerstone for understanding financial contagion. Their main claim is a paradox: financial networks can be **robust yet fragile**. This means that while the probability of contagion might be low, the events themselves are extremely severe when they occur.

4.1 Modeling the Agent: The Bank Balance Sheet

To understand contagion, we must first define the node. In the GK model, a bank's stability is determined by its balance sheet. A bank defaults if its capital buffers are insufficient to absorb the losses from its defaulted counterparties.

4.2 Connectivity and the Topology of Risk

The model focuses on the following key variables:

- **z (Average Degree):** The mean number of counterparties per bank.
- **Frequency (Y):** The probability that a single initial shock triggers a systemic crisis (defined as $> 5\%$ of banks failing).
- **Extent:** The magnitude or gravity of the contagion (total fraction of failed banks).

The simulation results reveal that contagion only exists within a specific domain of connectivity:

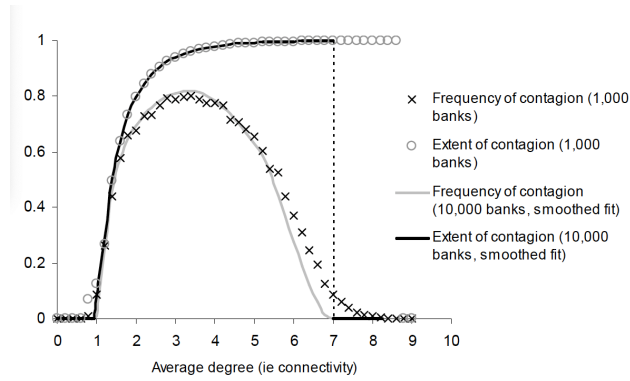


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1. **Low Connectivity ($z < 2$):** The network is too fractured. Shocks remain local because there are not enough links to allow propagation.
2. **The Danger Zone ($z \approx 3$):** The network is connected enough for a shock to propagate, but not diversified enough to absorb it. The frequency of contagion is maximal here.
3. **Strong Connectivity ($z > 5$):** Robustness improves. Banks share risk across many neighbors, allowing them to absorb small shocks without defaulting.
4. **The Boundary of Stability:** Beyond a certain threshold (often shown as a **dashed line**), contagion disappears in simulations. The apparent drop to zero in "Extent" plots doesn't mean crises become smaller; it means they stop happening altogether, making the extent undefined.

4.3 Non-Linearity and Indistinguishable Shocks

Risk in this model is strictly **non-linear**. Moving from 2 to 3 connections does not necessarily improve the safety of the network; in fact, it often increases the probability of a global collapse.

Furthermore, we observe a "knife-edge" property: two initial shocks that appear identical can lead to vastly different outcomes. For $z \approx 8$, a shock might have zero effect, or it might lead to the **total collapse** of every bank in the system. This makes the system extremely difficult to regulate using standard oversight tools.

4.4 The Physics of Contagion: Phase Transitions

As physics students, we recognize this behavior as a **phase transition**. Much like the transition between ice and water, where a microscopic change in temperature (one parameter)

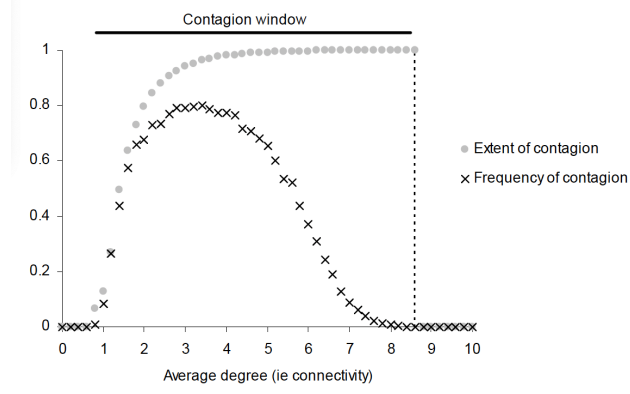


Figure 7: Enter Caption

leads to a macroscopic change in the state of the system, the financial network undergoes a structural shift.

The transition from a stable system to a collapsing one is not gradual; it is a sudden change in the global topology of defaults triggered by crossing a connectivity threshold.

4.5 Structural Convergence and the $N \rightarrow \infty$ Limit

A critical finding in our analysis is that the Gai-Kapadia mechanism is **structural, not numerical**. Because we have shown that the GK model is essentially Gaussian, we can apply the **Central Limit Theorem**.

When we compare simulations of 1,000 banks versus 10,000 banks, the results converge. This proves that the "robust yet fragile" nature is not an artifact of a small sample size or a specific simulation run, but a fundamental property of the network that holds as $N \rightarrow \infty$. The risk is embedded in the very geometry of the interbank links.

5 Comparison to Real Life: Mediocristan vs. Extremistan

While the Gai-Kapadia model provides a robust mathematical foundation, we must ask: how does it compare to the actual financial networks we observe today? Both models attempt to explain the "robust yet fragile" nature of the world, but they do so through entirely different mechanisms.

5.1 The Gaussian Perspective (Gai-Kapadia)

In a Gaussian world, the system is governed by the **Law of Large Numbers**. Risk is predictable and described by the mean.

- **Robustness via Diversification:** Stability comes from banks having many similar connections. By spreading risk thin, the system can absorb many small shocks.
- **Fragility via Magnitude:** The system breaks only when the total volume of shocks exceeds a critical threshold.
- **Phase Transition:** There is a sharp boundary between *Subcritical* (defaults are local and die out) and *Supercritical* (defaults reach a critical mass and trigger global contagion).

5.2 The Power-Law Perspective (Scale-Free Networks)

In the real world, interbank networks often follow a Power Law (scale-free) distribution. Here, the "average" bank is irrelevant; the system is defined by its **Hubs**.

- **Robustness via Hubs:** These massive nodes (the JP Morgans or Goldman Sachs of the world) act as stabilizers. They can absorb thousands of small shocks from the periphery without flinching.
- **Fragility via Hub Failure:** The system has an "Achilles Heel." While it is immune to random failures, the collapse of a single hub dominates everything else.
- **The Constant Supercritical Phase:** Unlike the Gaussian world, the supercritical phase in power laws is often **always present**. Because hubs are deeply embedded in the structure, the potential for a "path" to a global crisis is never zero.

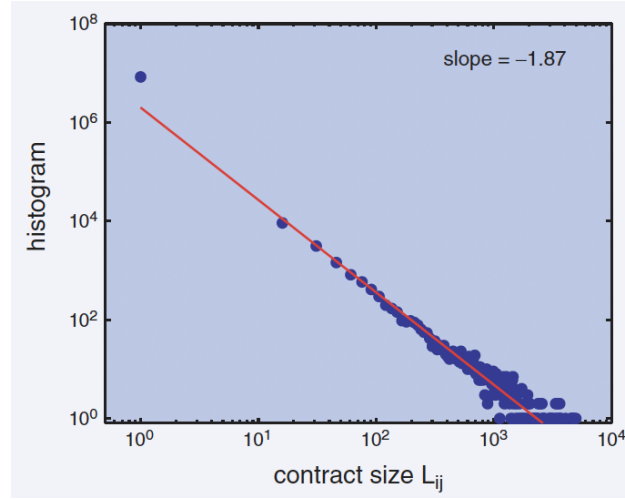


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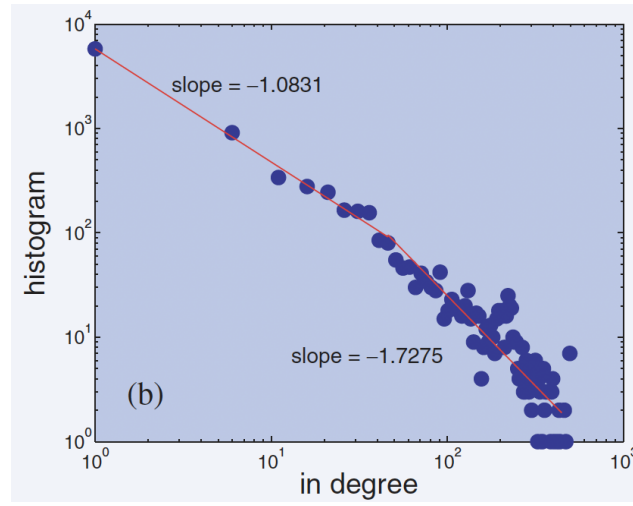


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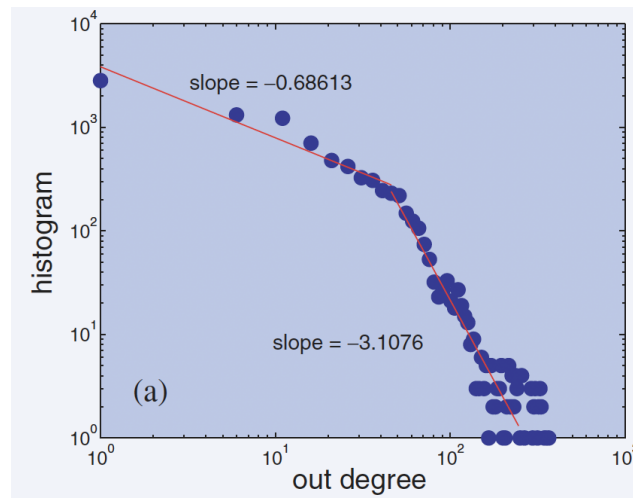


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Feature	Gaussian World (GK)	Power-Law World
Robustness	Diversification (Safety in numbers)	Hubs (Stability via massive centers)
Fragility	Critical magnitude (Shock is too large)	Targeted hub failure (A few exposures dominate)
Shocks	Shocks are stabilized by spreading them	Shocks on hubs are amplified; shocks on periphery are absorbed
Physics Analogy	Sharp Phase Transition (Ice to Water)	Criticality as a Structural Property

Table 2: Structural differences in systemic risk perception.

5.3 Summary Comparison

5.4 The Final Takeaway

The most important lesson for us to remember is that in a **Gaussian world**, risk is described by the mean. In a **Power-Law world**, risk is dominated by the extremes.

Systemic risk is not simply a "large version" of an average shock; it is a structural property of the network itself. If we only look at the "average" bank, we are blind to the reality that the system's fate rests on the tails of the distribution.

Note: For a deeper visual intuition on why these power laws are so pervasive and counter-intuitive, we highly recommend the Veritasium coverage on "The 80/20 Rule" and the math of networks. It perfectly illustrates how "Fairness" and "Normalcy" are rarely the default state of complex systems.

6 Conclusion and Outlook

6.1 Final Synthesis: The Connectivity Paradox

For decades, the prevailing economic intuition was that increasing connectivity would necessarily make the financial system more stable through better risk-sharing. Our analysis of the Gai-Kapadia model and Power-Law dynamics challenges this simplistic view.

While connectivity improves stability **locally** by allowing banks to diversify their exposures, it creates **systemic fragility** once certain topological thresholds are crossed. This highlights a critical lesson: economic mechanisms rarely behave in a linear fashion. More connectivity does not always imply more stability; instead, it can create "super-highways" for contagion.

6.2 Key Policy Implications

The results of this study suggest that monetary and regulatory policies must shift their focus from individual health to **network health**:

- **Network Supervision:** It is not enough to monitor a bank's internal balance sheet. Regulators must supervise the *structure* of the network to prevent the formation of dangerous contagion channels.
- **Accounting for Thresholds:** Policy-making must account for non-linear effects. A small change in interbank interest rates or lending requirements can push the system across a phase transition boundary, moving it from a stable state to a supercritical one.

6.3 Outlook: The Reality of Capital Buffers

During our research, Clément and I questioned the realism of the model's parameters, specifically the 4% capital buffer often used in theoretical simulations. Looking back at the 2008 crisis, the case of **Lehman Brothers** is particularly telling.

While the exact numbers vary across different types of assets, the systemic failure of 2008 proved that the existing buffers were insufficient to stop the "domino effect." This led to the implementation of **Basel III**, which significantly tightened regulations:

- **Pre-2008/Basel II:** Total capital requirements were often around 8%.
- **Post-Crisis/Basel III:** These were increased to roughly 13% (including conservation and systemic risk buffers).

This shift in the "real world" mirrors what we see in the model: to move the system away from the "fragile" zone, you must either decrease the connectivity (de-leveraging) or increase the "melting point" of the nodes (higher capital). Basel III chose the latter.

Ultimately, as physics students, we see that the financial system is a **complex dynamical system**. To manage it effectively, we must stop treating it as a collection of parts and start treating it as a living, interconnected network.

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A Annex: Mathematical Foundations of the Gaussian Regime

A.1 Formal Proof of the Gaussian Limit in Gai-Kapadia (GK)

In the GK model, the vulnerability of the system is often masked by the assumption of finite moments. Let us formalize the proof that losses in this regime are self-averaging.

1. Structural Assumptions:

- The network is a random Erdős-Rényi graph where the degree k_i follows a Poisson distribution: $k_i \sim \text{Pois}(\lambda)$.
- The interbank exposures w_{ij} are i.i.d. with finite mean μ and finite variance σ^2 .
- The loss L_i for bank i is the sum of exposures to defaulted neighbors.

2. The Loss Equation: Assuming a fraction q of neighbors default, the loss is:

$$L_i = \sum_{j=1}^{k_i} X_j, \quad \text{where } X_j = w_{ij} \cdot \mathbf{1}_{\{j \text{ defaults}\}}$$

Since the number of terms k_i is itself a random variable, L_i is a **compound random variable**.

3. Wald's Identity (The Physics Perspective): To find the total variance of the system's stress, we apply Wald's Identity. The expected loss and variance are:

$$E[L_i] = E[k_i]E[X] = \lambda(q\mu)$$

$$\text{Var}(L_i) = E[k_i]\text{Var}(X) + (E[X])^2\text{Var}(k_i)$$

Substituting the Poisson property ($\text{Var}(k_i) = E[k_i] = \lambda$):

$$\text{Var}(L_i) = \lambda [\text{Var}(X) + (q\mu)^2]$$

4. Convergence: As the network size $N \rightarrow \infty$ and the average degree λ becomes large, the Central Limit Theorem (CLT) dictates:

$$\frac{L_i - E[L_i]}{\sqrt{\text{Var}(L_i)}} \xrightarrow{d} \mathcal{N}(0, 1)$$

Conclusion: In the GK framework, the probability of an extreme loss (a tail event) decays **exponentially** as e^{-x^2} . This means the model structurally ignores the possibility of "Black Swan" events.

A.2 When the CLT Fails: The Power-Law Regime

If we replace the Gaussian assumption with a Power-Law (Pareto) distribution for exposures $w_{ij} \sim x^{-\alpha}$ with $\alpha < 2$, the variance becomes **infinite**.

In this case, the standard CLT no longer applies. Instead, the sum of losses converges to a **Lévy Alpha-Stable Distribution**.

- **Gaussian World:** $L_{total} \approx \sum L_i \rightarrow$ stable mean (shocks cancel out).
- **Power-Law World:** $L_{total} \approx \max(L_i) \rightarrow$ the largest single shock dominates the entire system.

This mathematical shift explains why a single Lehman Brothers could collapse a global system that was "statistically stable" under Gaussian assumptions.